Teaching Notes

Creating SQL Server Databases using Visual Studio

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MacBook and Chromebook users are advised that this Teaching Note may not be applicable.

You may use Visual Studio Professional 2019 pre-installed on lab PCs on campus. You can also download and install Visual Studio Community 2022 on your PC. When you install Visual Studio Community 2022, please make sure you select the "Data storage and process" package shown in Figure 1.

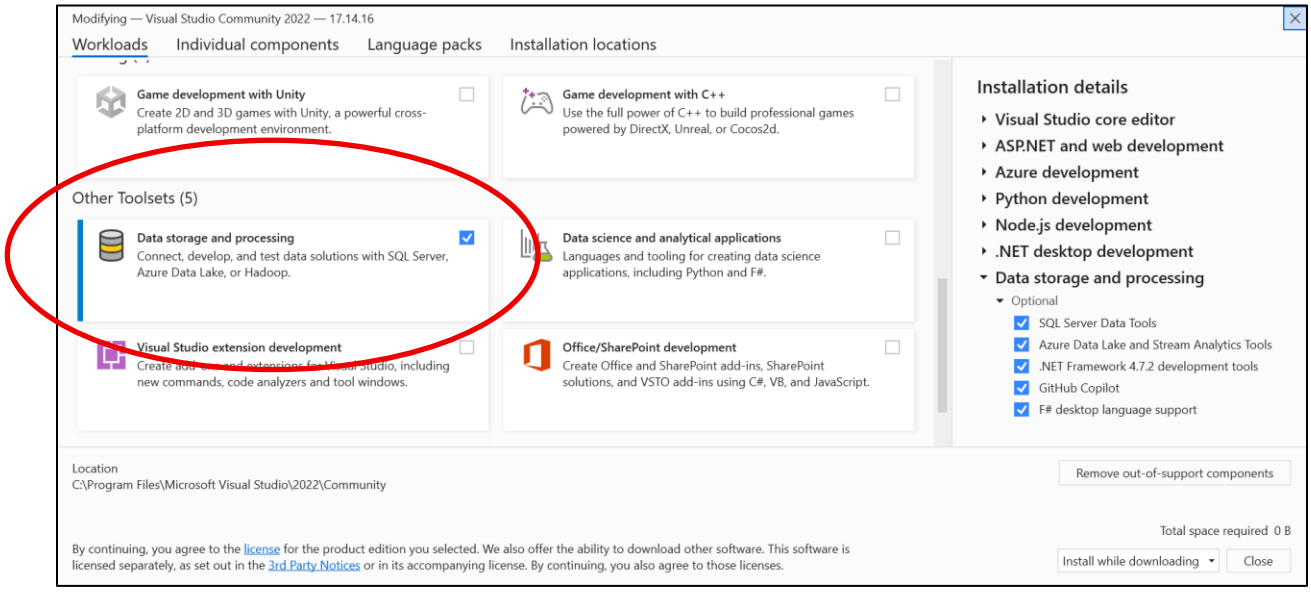

Figure 1

If you have already installed Visual Studio Community 2022 without the "Data storage and processing" package, you may modify the installation configuration by running the Visual Studio Installer program, as shown in Figure 2.

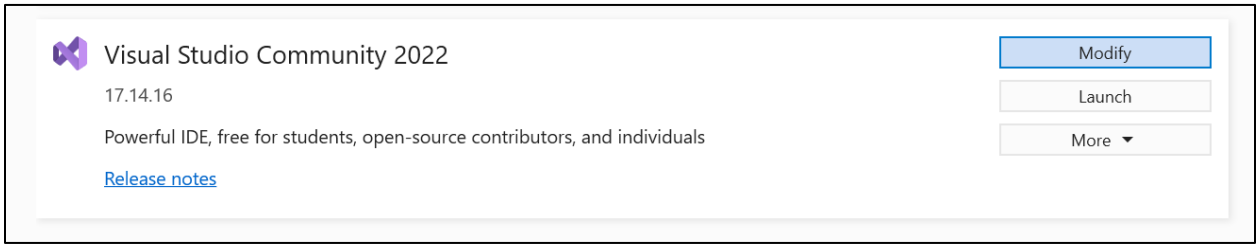

Figure 2

Once you launch Visual Studio 2022, you can create a new SQL Server database project as shown in Figure 3 and Figure 4.

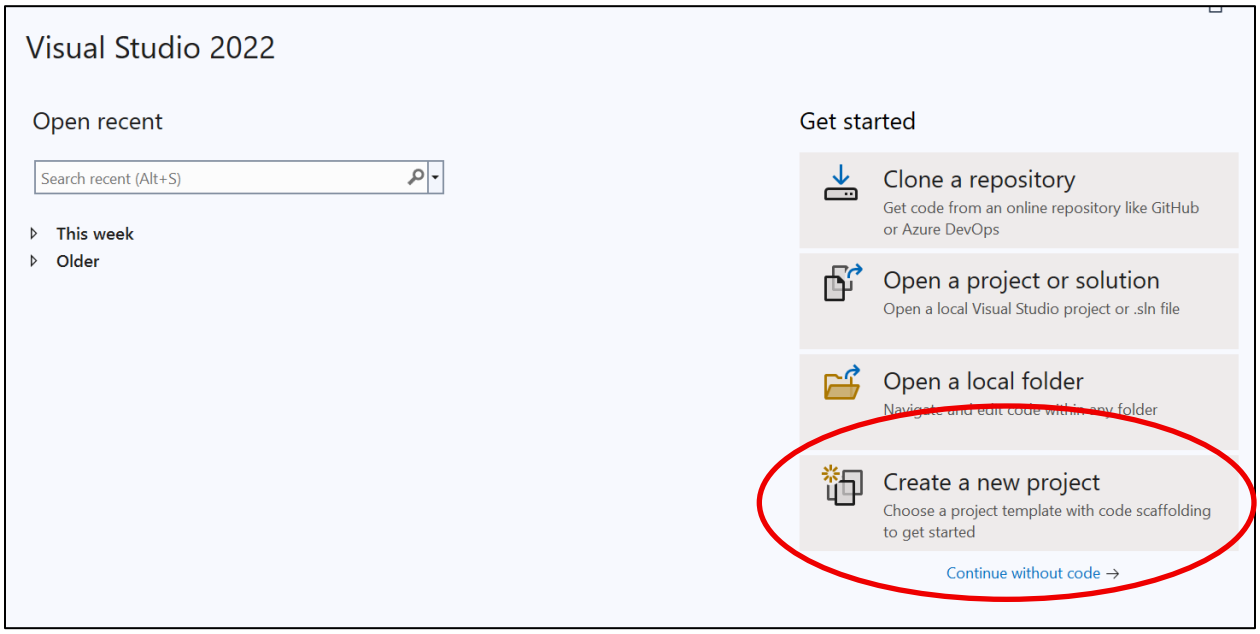

Figure 3

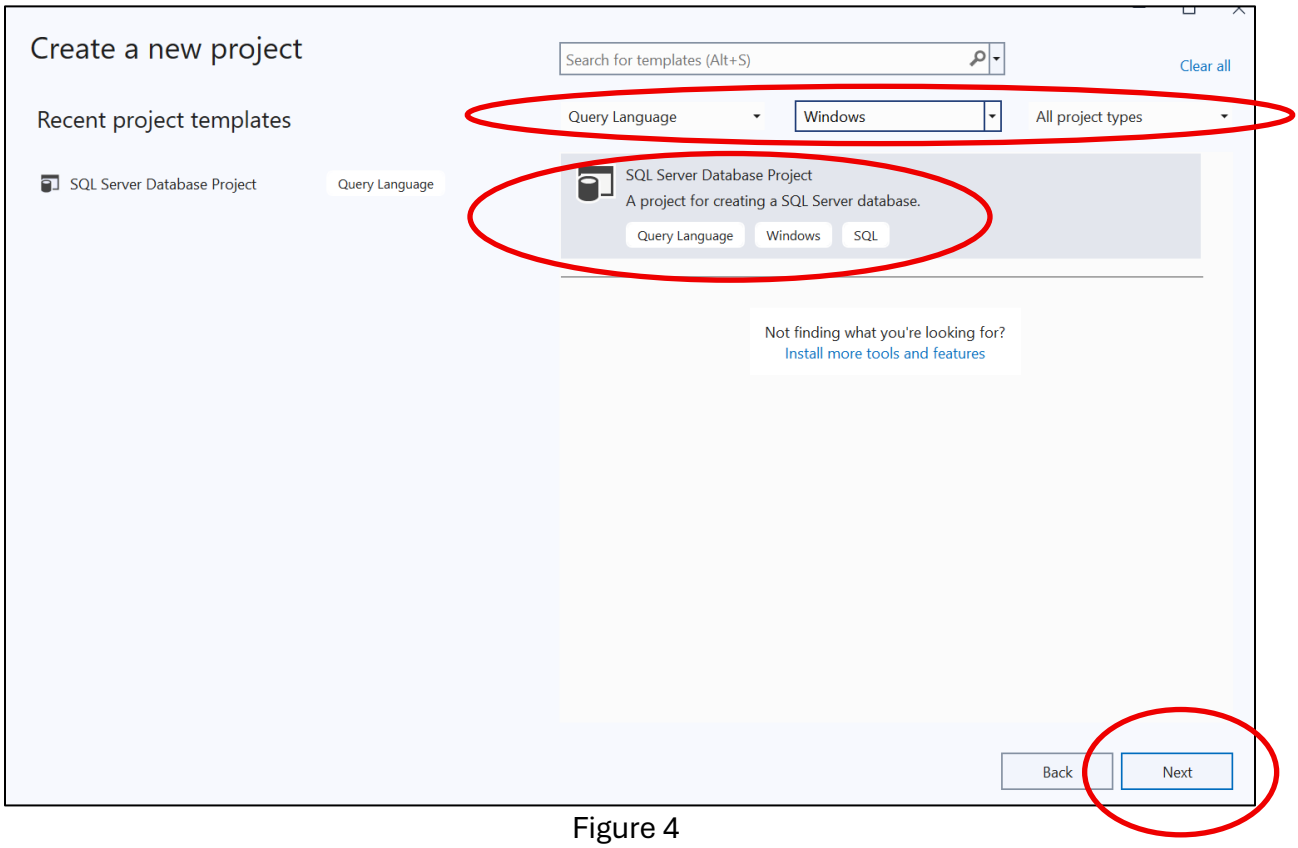

Figure 4

In Visual Studio, a *project* is contained in a *solution*. A solution serves as a container that organizes one or more related projects, along with associated configurations and any miscellaneous files that are not tied to a specific project. Generally, a solution corresponds to a file folder, and a project corresponds to a subfolder inside the solution folder. You may use the same names for both the solution folder and the project folder when creating a new database project, as shown in Figure 5.

When you create a project, Visual Studio saves the corresponding solution/project to the default location *"%USERNAME%\source\repos"*. You may change the solution and project folders' location on your PC by clicking the highlighted button in Figure 5.

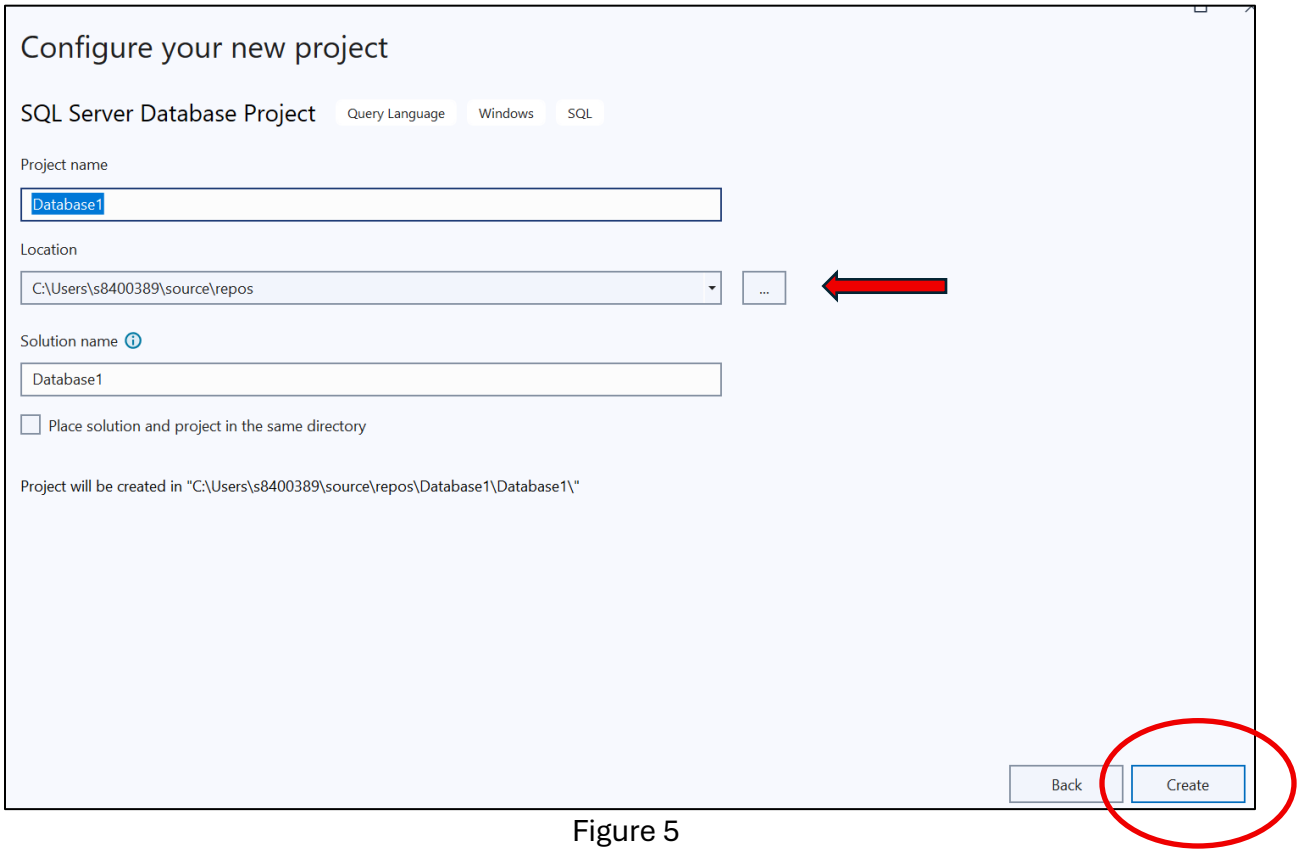

Figure 5

Once you create a new database project, as shown in Figure 6, your solution and project will appear in the "Solution Explorer" window, and your project will also appear in the "SQL Server Object Explorer" window.

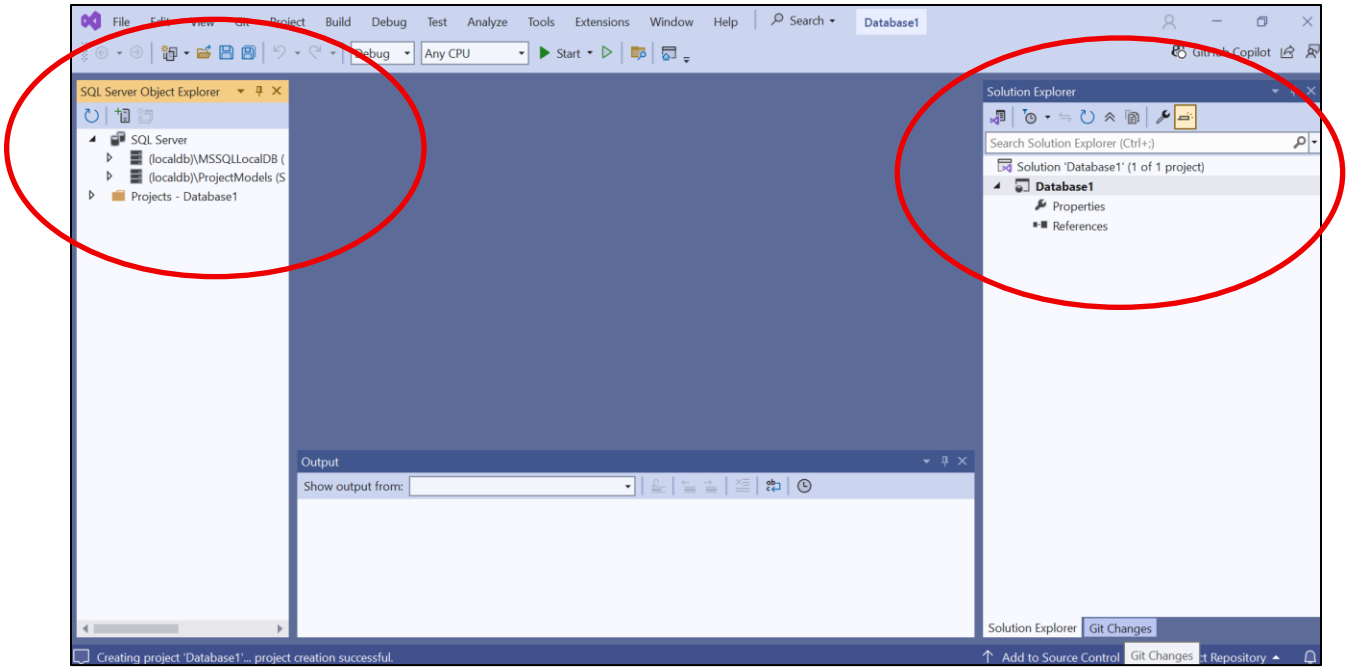

Figure 6

You need both the “Solution Explorer” and “SQL Server Object Explorer” windows for creating a database. If you mistakenly close either, you can display it by clicking **[View]** and then selecting the window, as depicted in Figure 7.

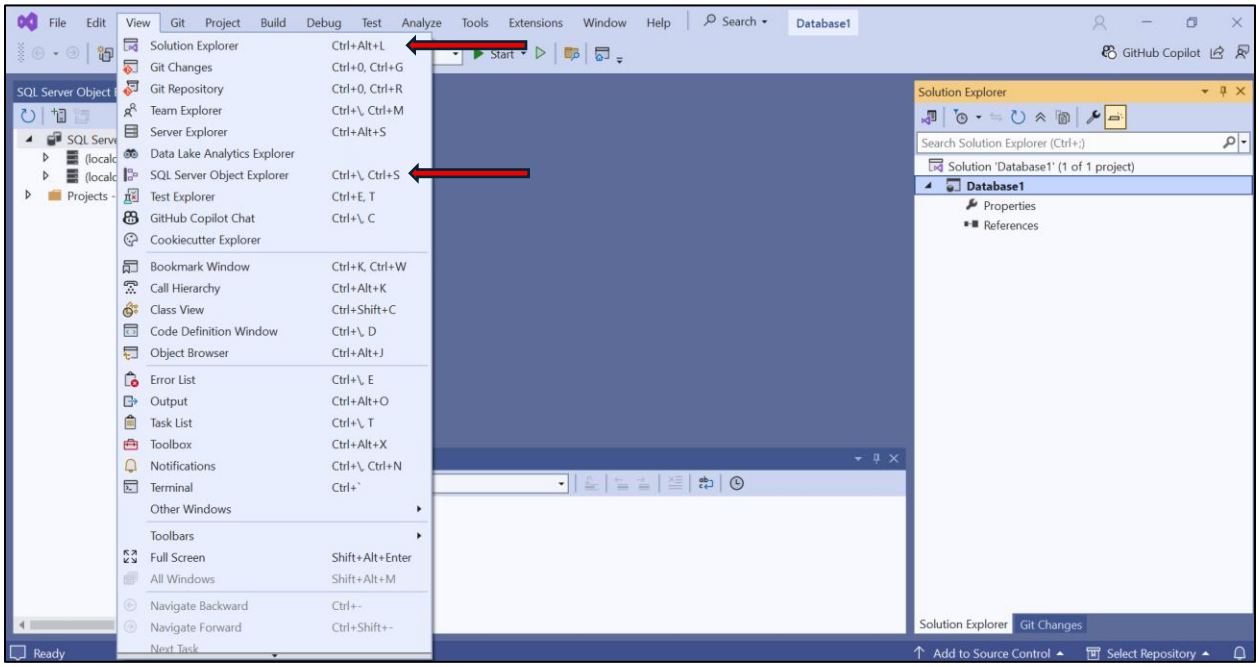

Figure 7

You can create database tables in a project by right-clicking the project name in the "Solution Explorer" window, then pointing to **[Add]** and clicking **[Table]**, as depicted in Figure 8. When the "Add New Item" window appears, as shown in Figure 9, <u>you must enter the name of the database table to be created (**do not** use the default table name)</u>, and then click **[Add]**.

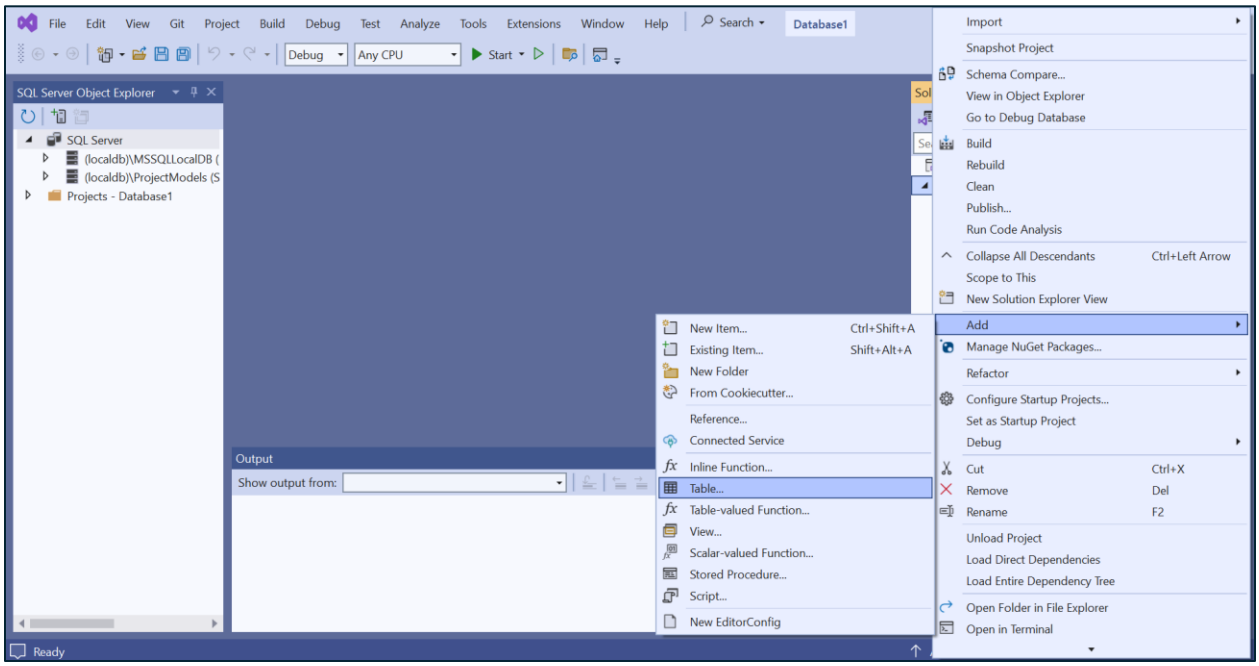

Figure 8

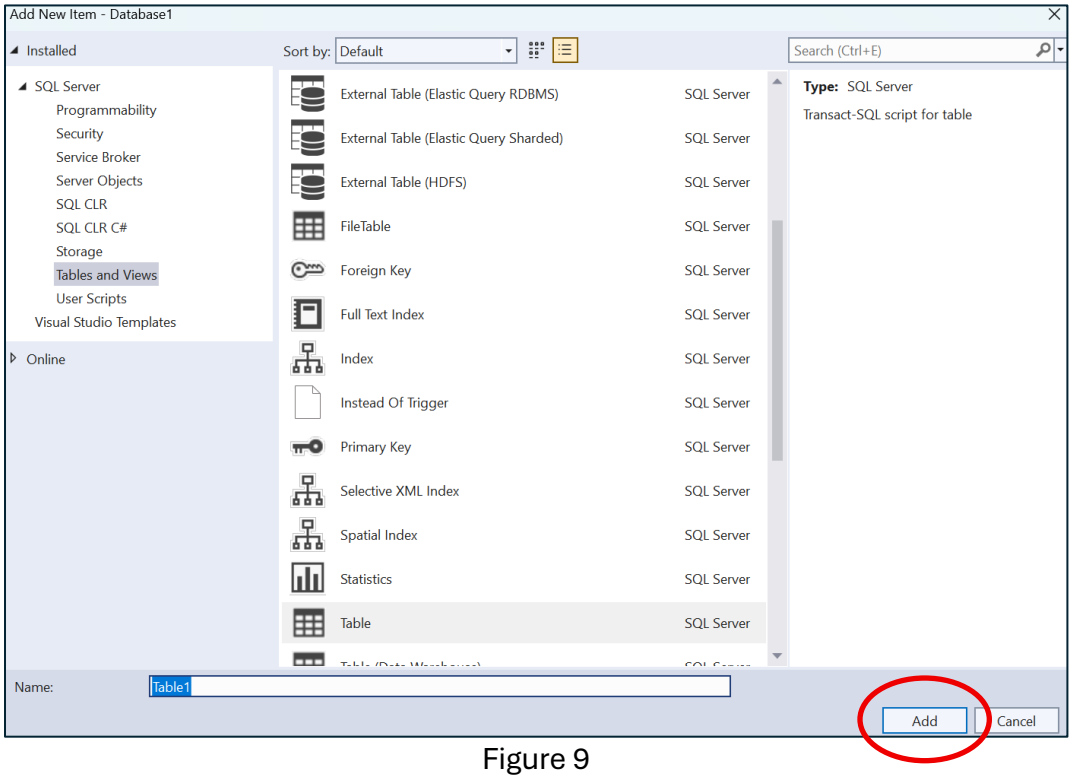

Figure 9

In the “Database Table Design” window in Figure 10, you can design the table, and the corresponding SQL statement will be automatically generated.

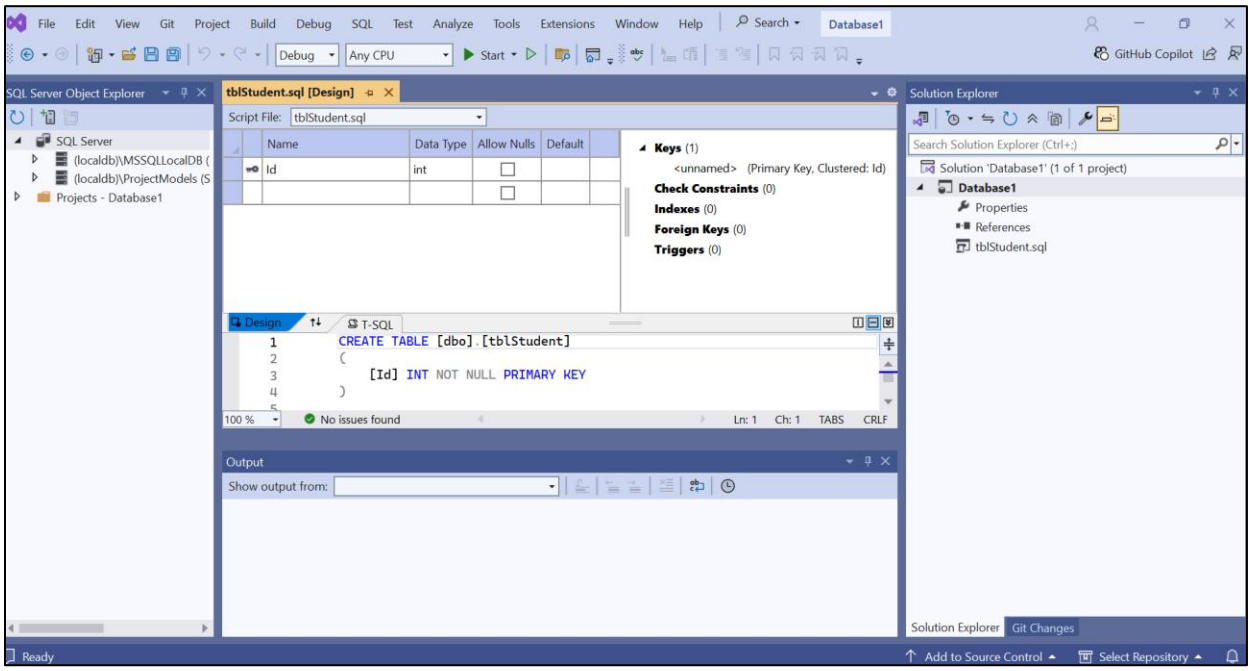

Figure 10

As shown in Figure 11, you can change the attribute names and their data types in the “Database Table Design” window.

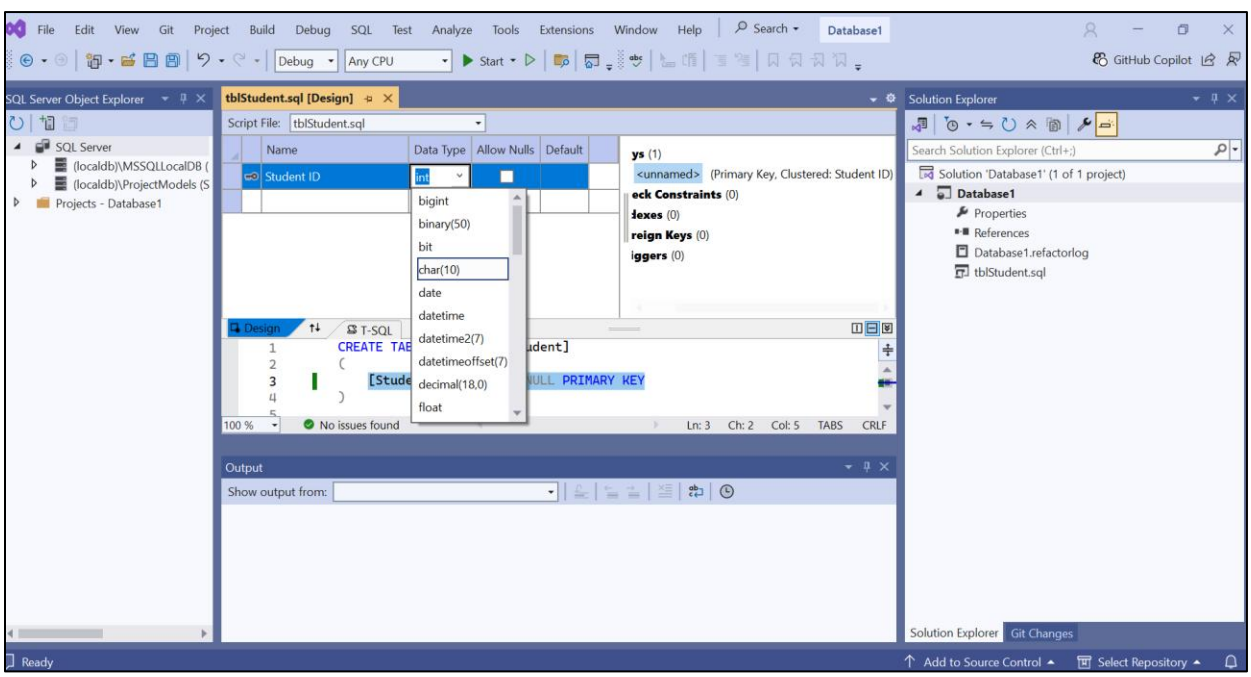

Figure 11

Figures 12 through 14 show the “tblStudent”, “tblCourse”, “tblOffice” tables, and the corresponding automatically generated SQL statements.

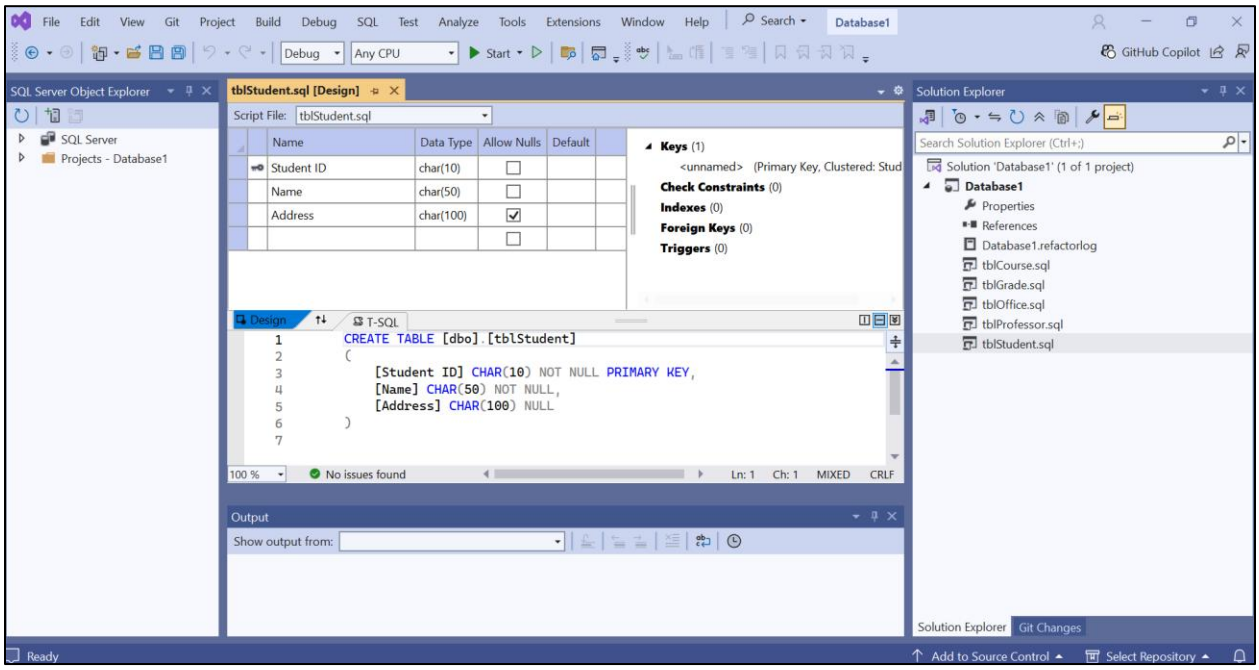

Figure 12

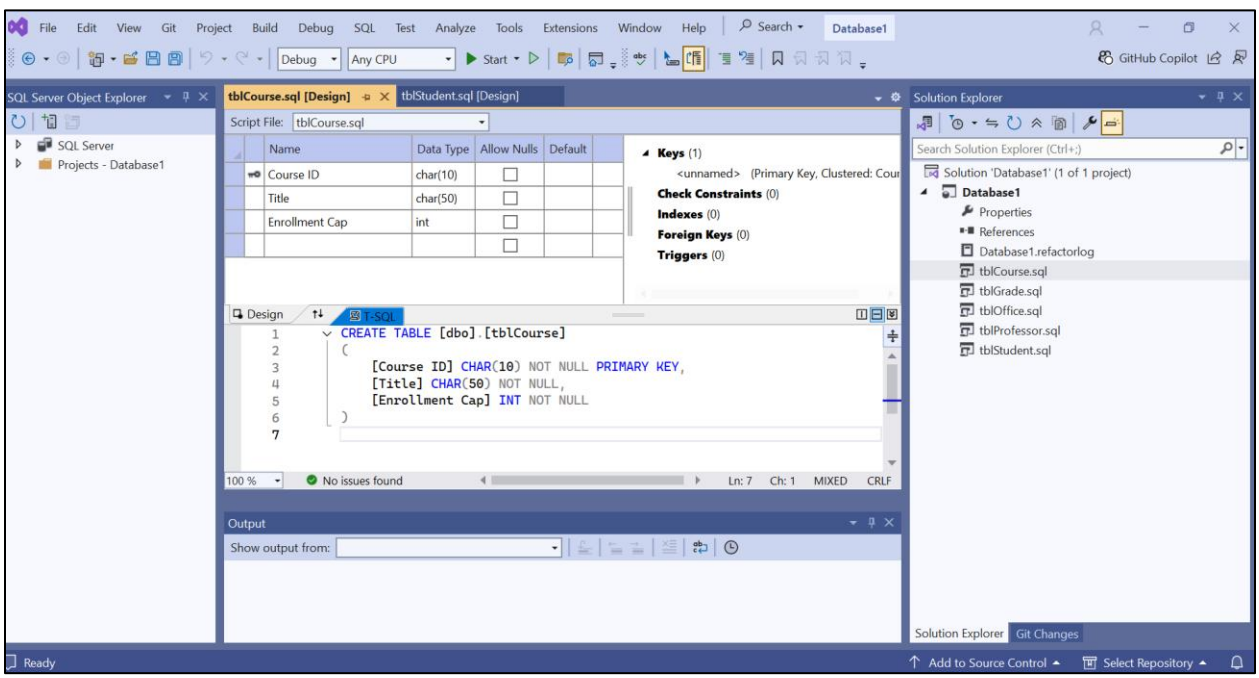

Figure 13

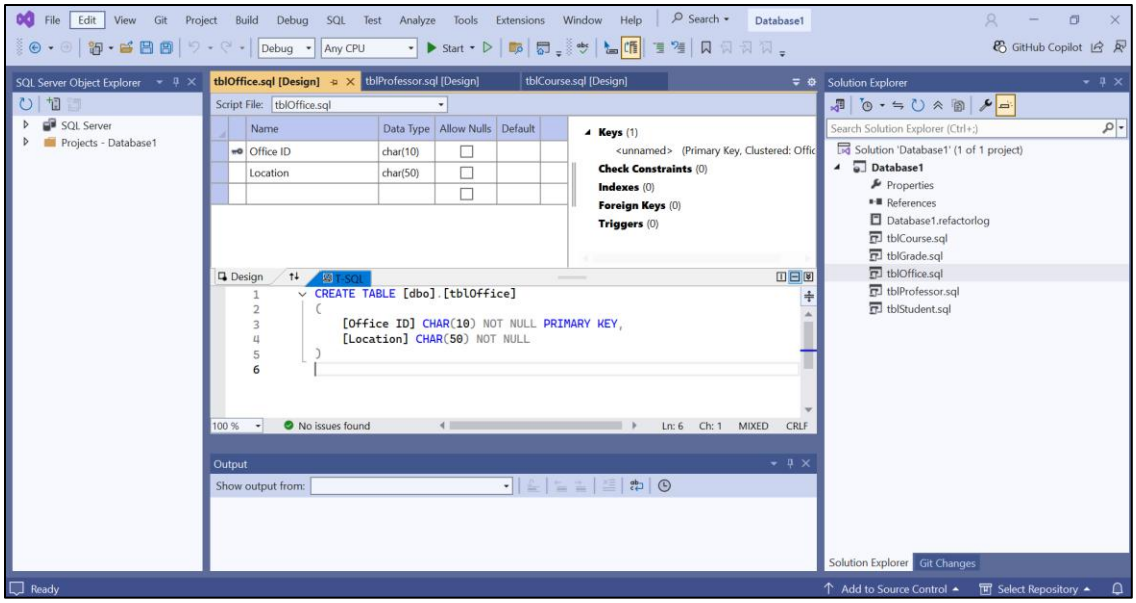

Figure 14

Figure 15 shows the "tblProfessor" table and the corresponding SQL statement. SQL statements are case insensitive, which means upper-case and lower-case letters are treated as equivalent. You must add

FOREIGN KEY ([Office ID]) REFERENCES tblOffice ([Office ID])

in the SQL statement in order to establish a binary relationship between the "tblProfessor" and "tblOffice" tables. In order to make the binary relationship a 1:1 binary relationship, you must add the keyword **UNIQUE** right after the data type of the attribute [Office ID] in the SQL statement.

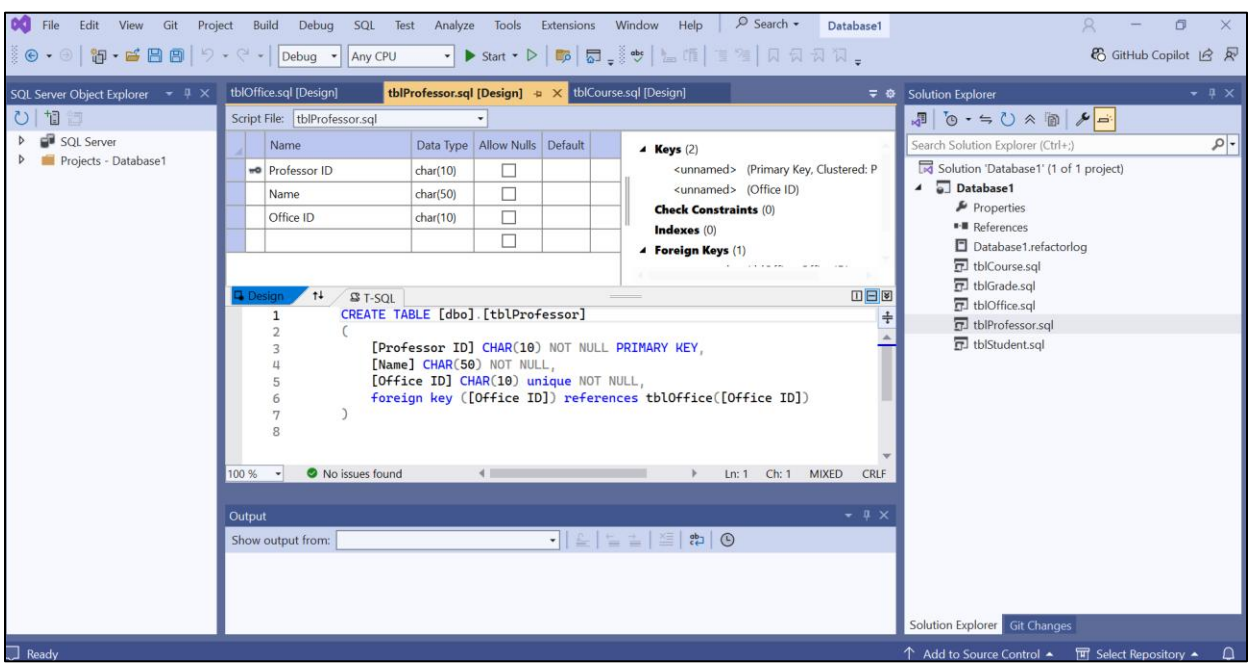

Figure 15

Figure 16 shows the design of the “tblGrade” table before creating its composite primary key. To create a composite primary, you click the button left of the key symbol (highlighted by the red arrow sign), drag to all three attributes, and then right-click. A menu will appear, as shown in Figure 17, to allow you to set all three attributes as the composite primary key. Figure 18 shows the design of the “tblGrade” table after creating the composite primary key.

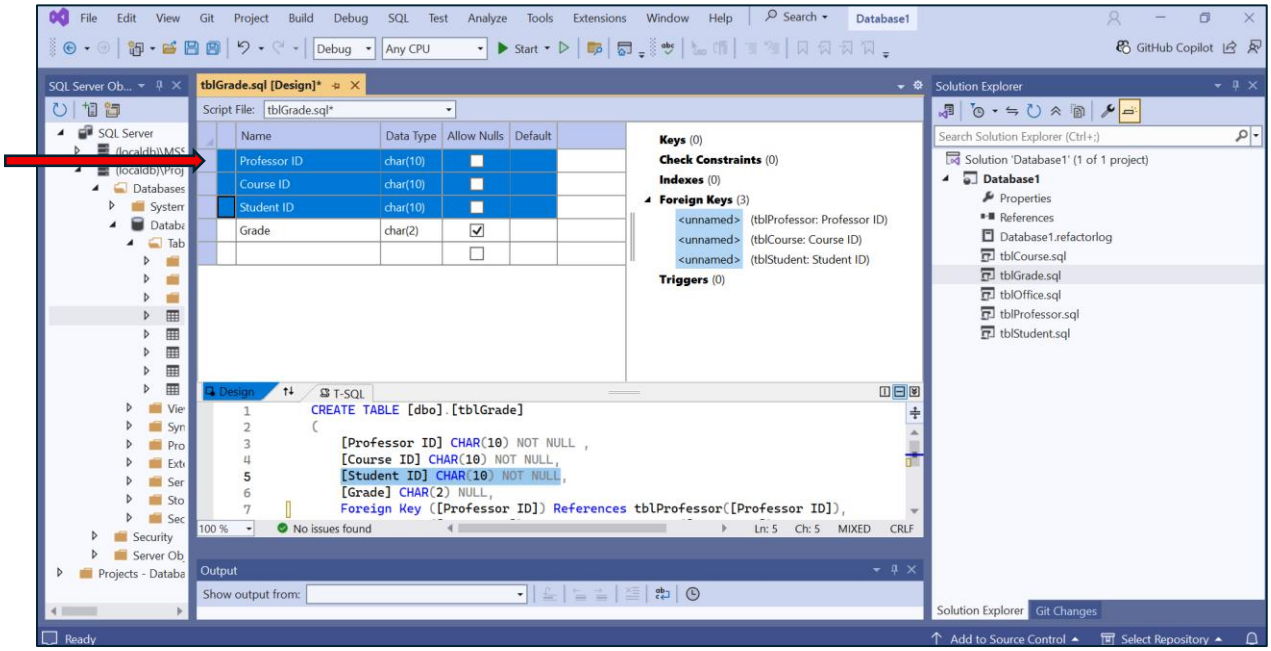

Figure 16

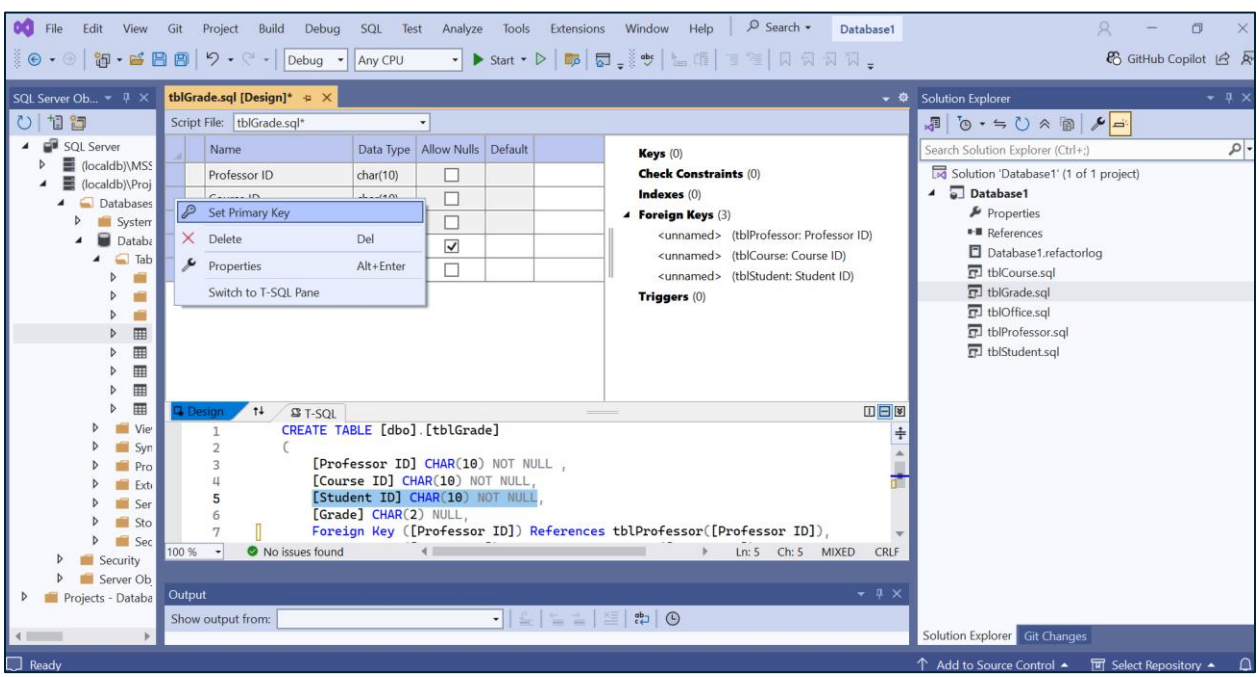

Figure 17

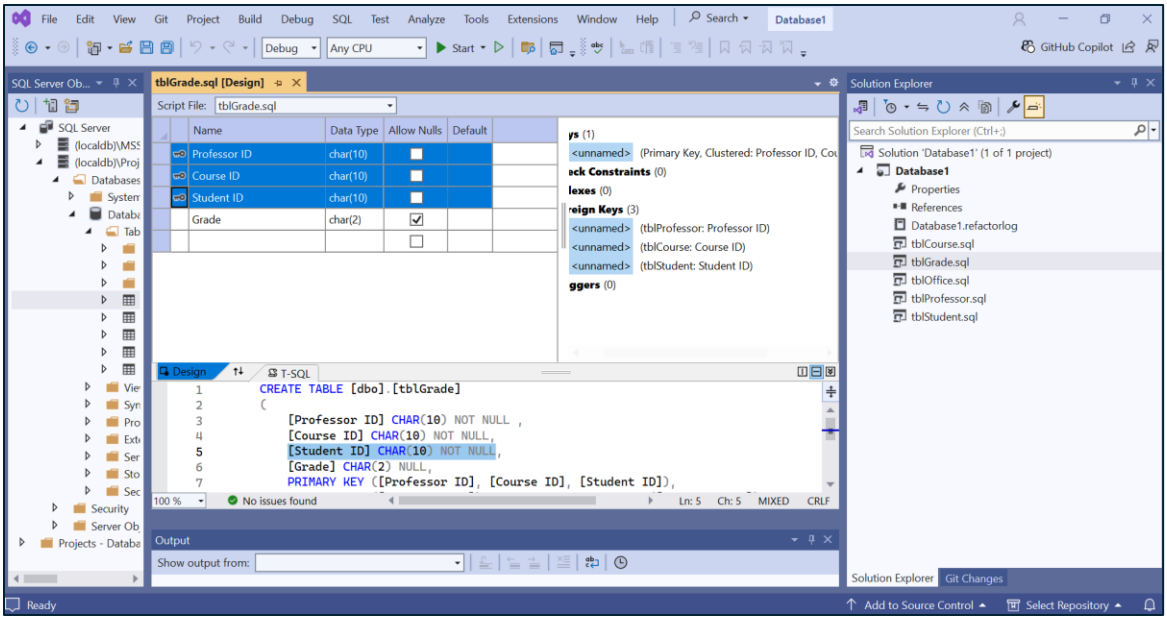

Figure 18

As shown in Figure 19, you must add

FOREIGN KEY ([Professor ID]) REFERENCES tblProfessor ([Professor ID]),
FOREIGN KEY ([Student ID]) REFERENCES tblStudent ([Student ID]),
FOREIGN KEY ([Course ID]) REFERENCES tblCourse ([Office ID])

in the SQL statement in order to establish the three binary 1:M relationships between the "tblGrade" table and the "tblProfessor", "tblStudent", "tblCourse" tables. After you finish creating all tables, you must save them.

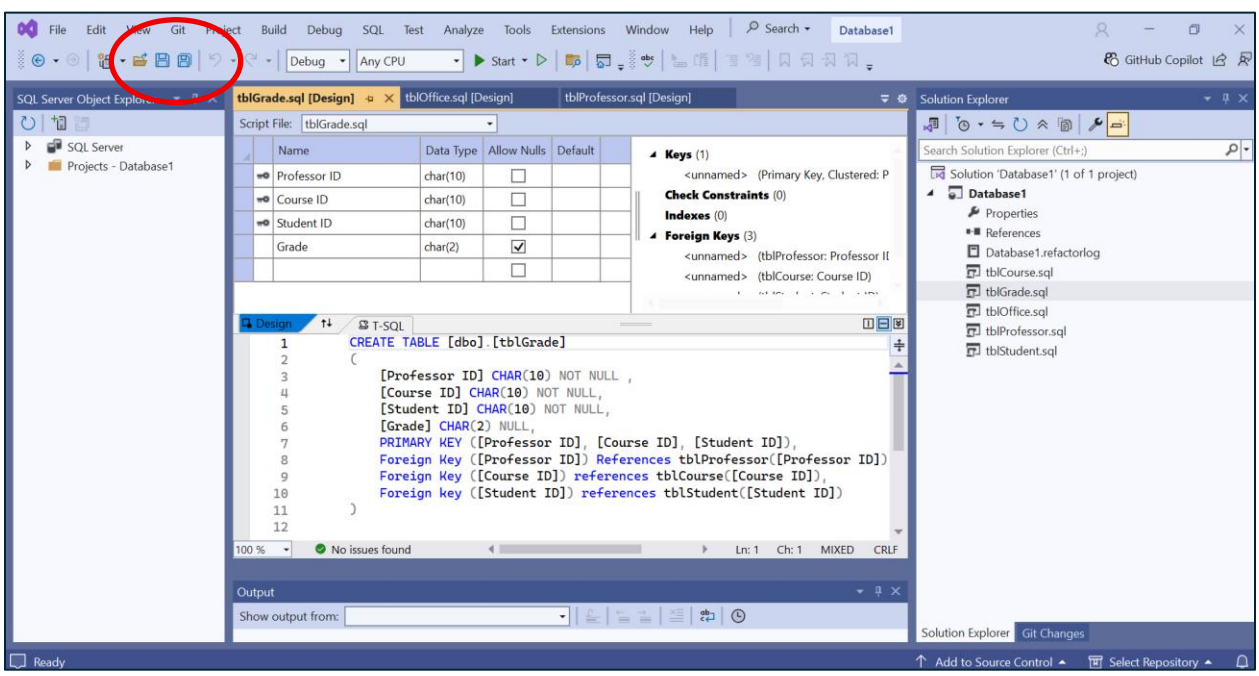

Figure 19

You must set up the SQL Server for your database project before you can build and deploy the database. The word “build” means to check the correctness of all SQL statements for creating all tables and then translate SQL statements into machine-executable code. The word “deploy” means to actually create all database files on the hard disk of your computer. For Visual Studio 2022, the SQL Server should be “SQL Server 2019”. For Visual Studio 2019, the SQL Server should be “SQL Server 2016”.

To set up the SQL Server, you right-click the “Properties” file in the “Solution Explorer” window and then click **[Open]**, as depicted in Figure 20. Then you select the right SQL Server as shown in Figure 21.

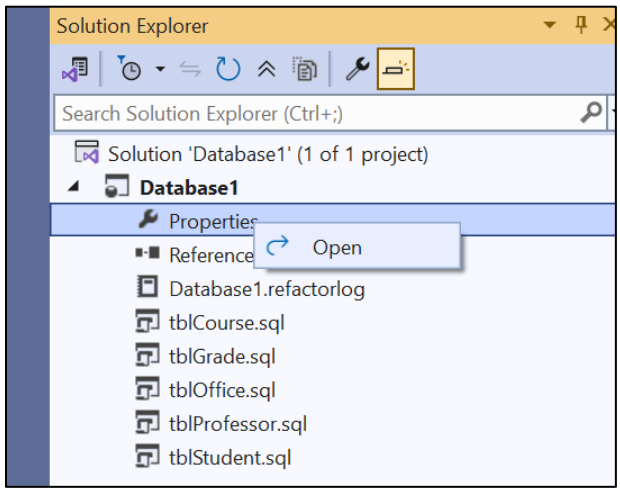

Figure 20

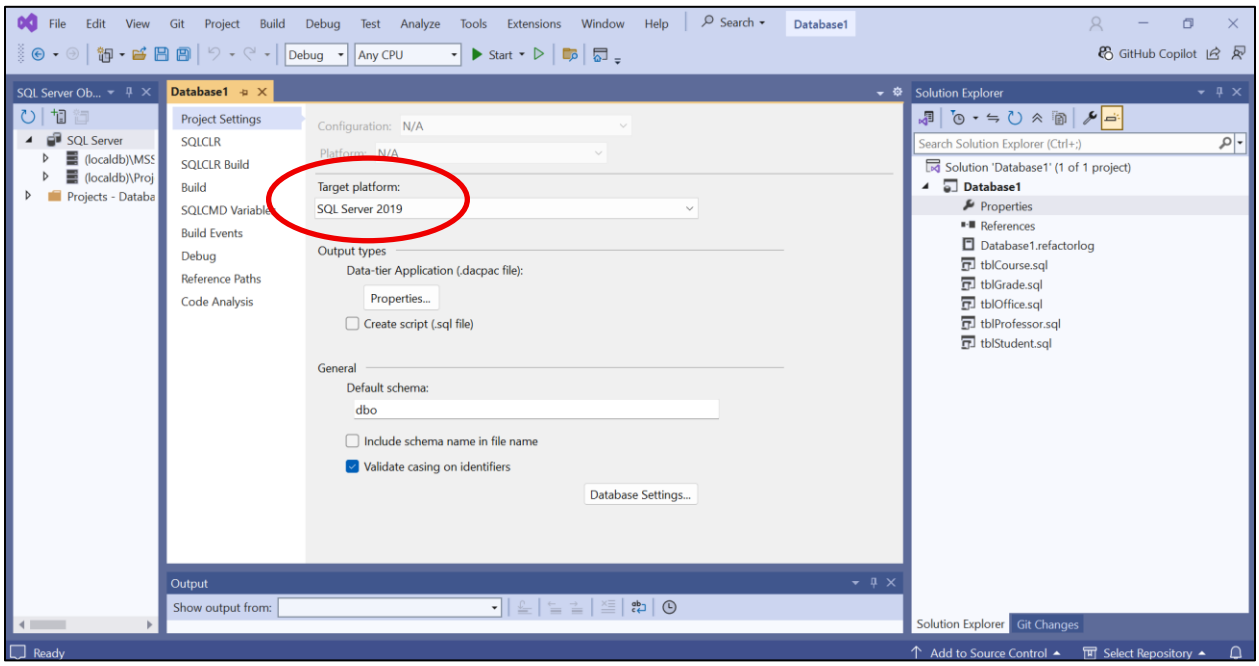

Figure 21

As shown in Figure 22, you click the **[Start]** button to build and deploy the database. If both build and deploy are successful, you will be able to see the messages as well as all tables in the “SQL Server Object Explorer” window.

To enter the data into a particular table, you find the table name under the second “localdb” in the “SQL Server Object Explorer” window, right-click the table, and then click **[View Data]**, as depicted in Figure 22. As shown in Figure 23, the “Database Table Data” window will appear to allow you to enter data into the table.

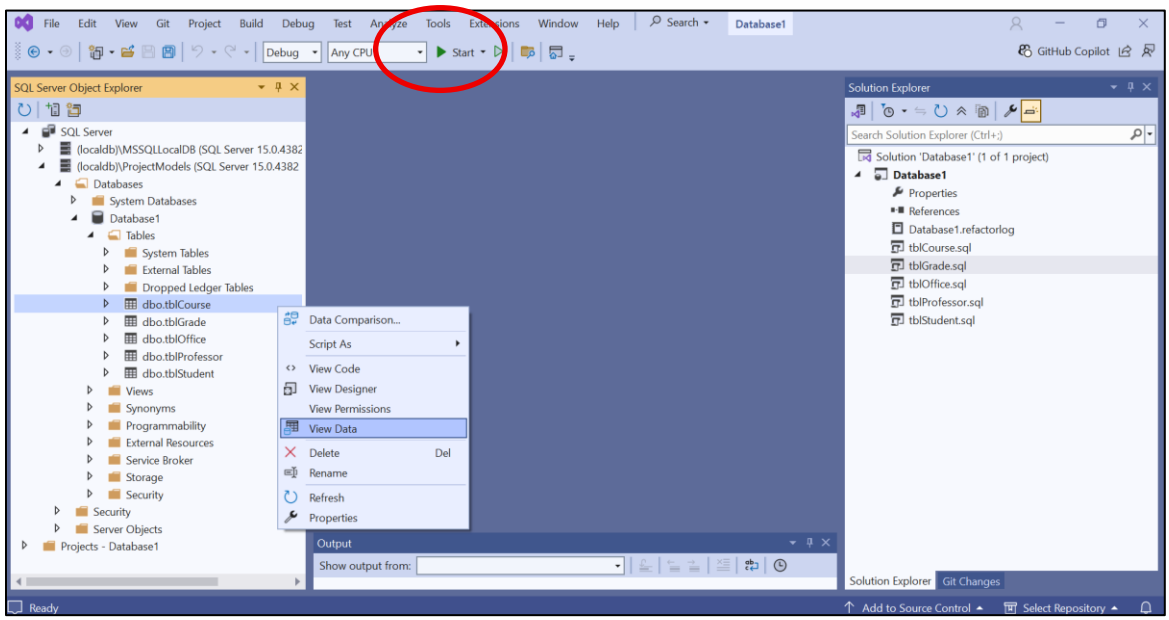

Figure 22

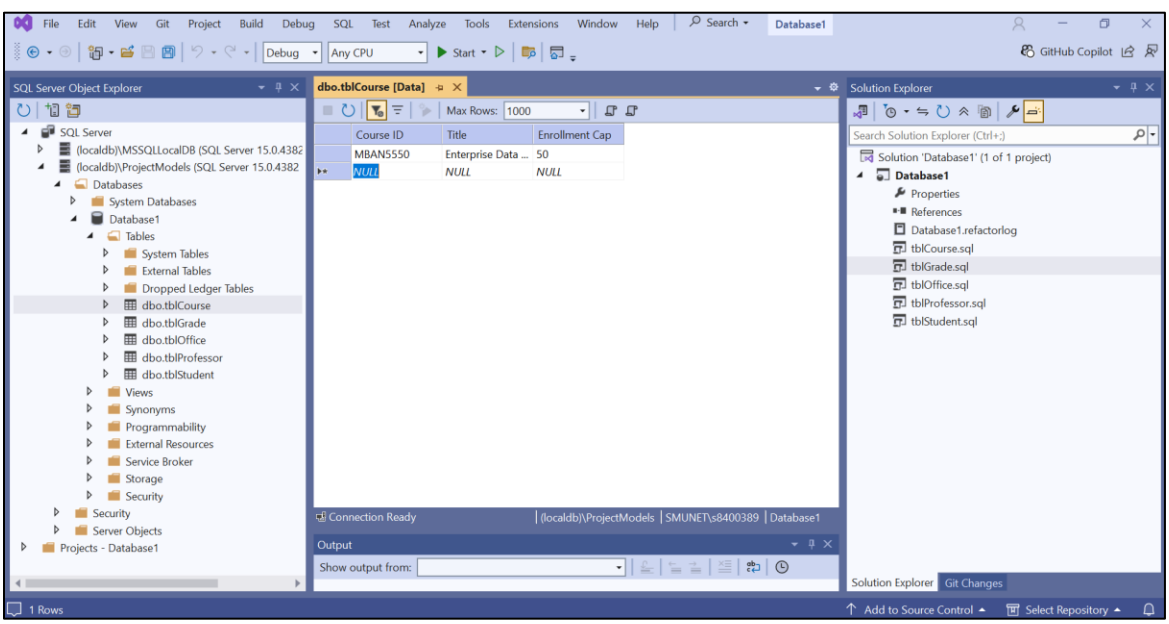

Figure 23

To add SQL statements for querying the data in the database, you right-click the database project name in the “Solution Explorer” window, then point to **[Add]** and click **[Script]**, as depicted in Figure 24.

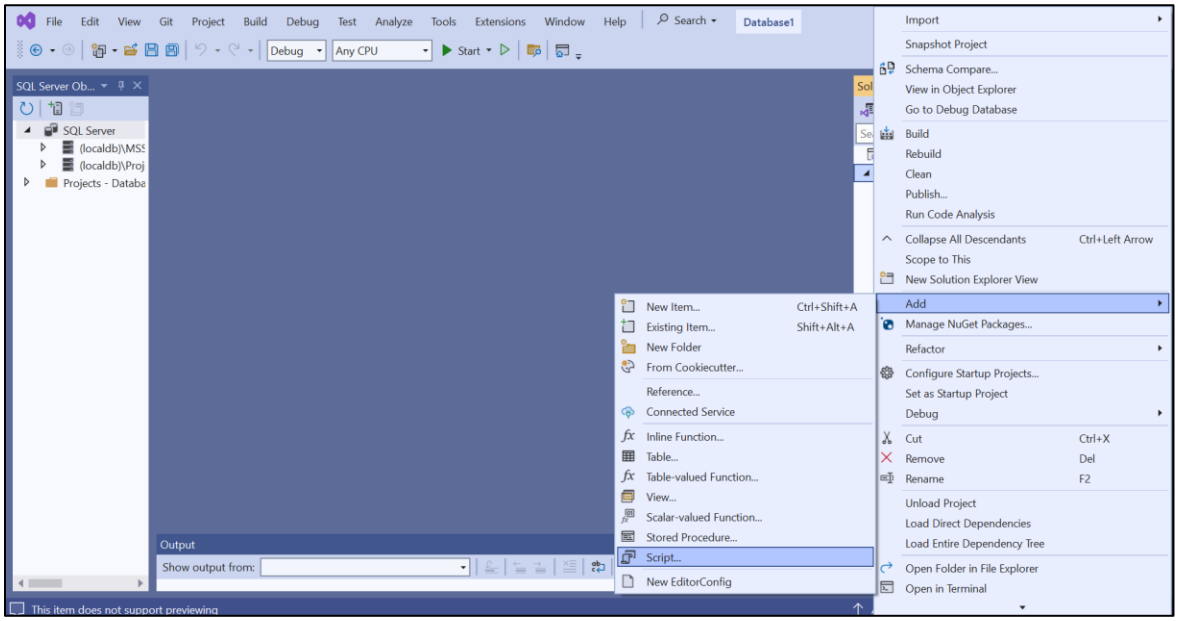

Figure 24

When the “Add New Item” window appears, as shown in Figure 25, you select **[Post Deployment Script]** and then click **[Add]**.

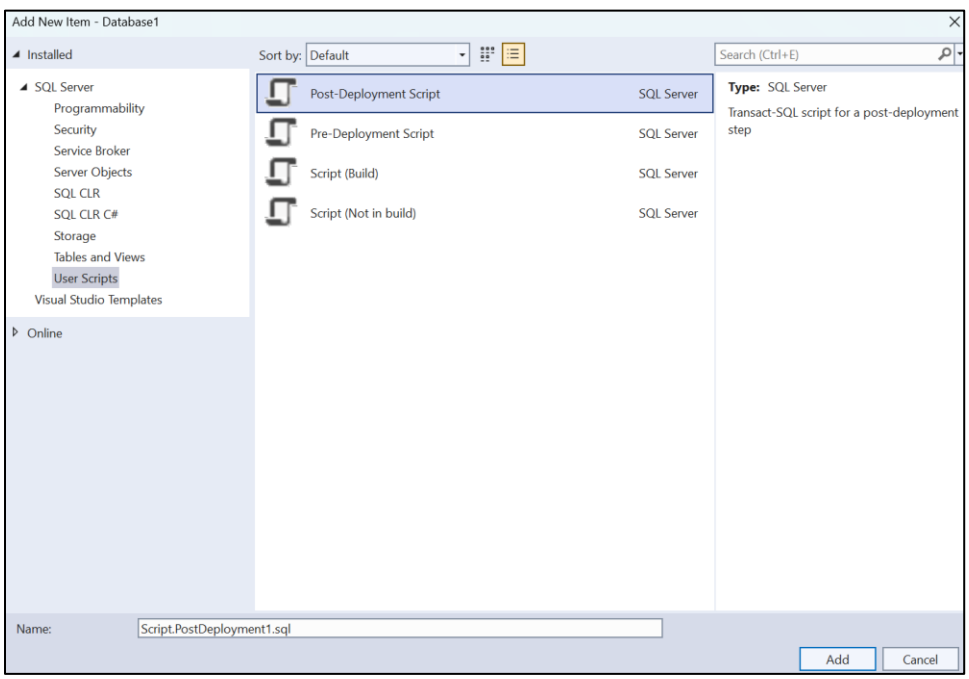

Figure 25

As shown in Figure 26, you can add the SQL statements to the script file, and then execute them by clicking the **[Execute]** button.

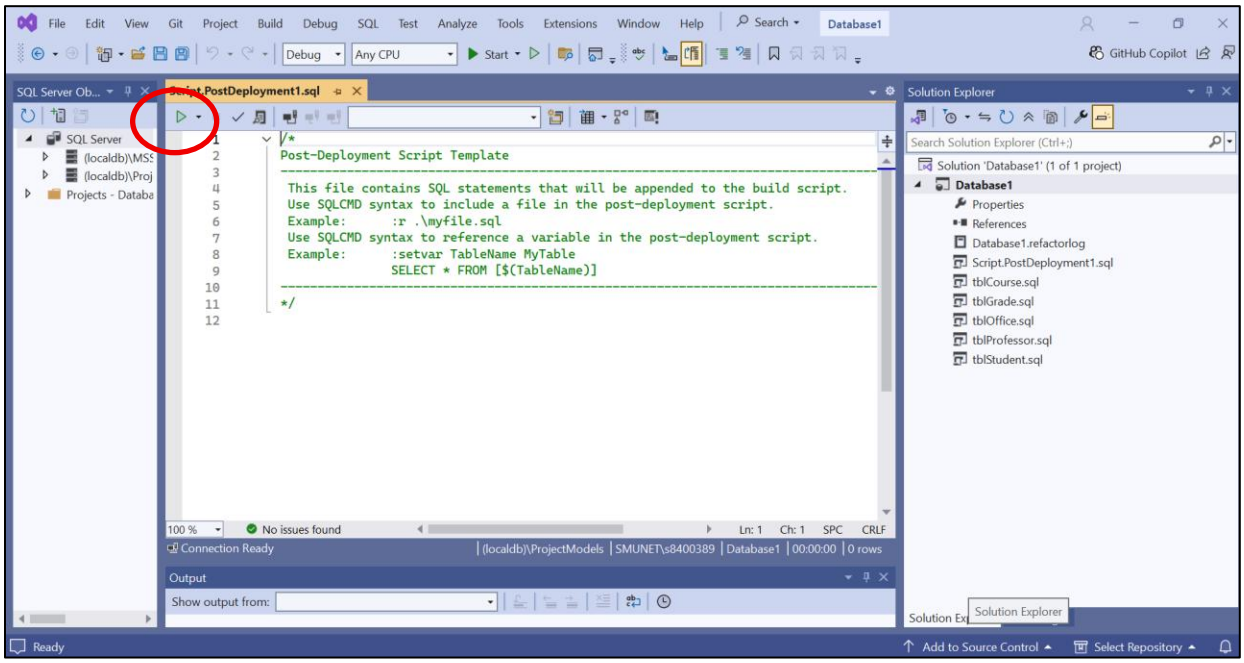

Figure 26